

RESEARCH ARTICLE

A novel method for predicting ecological interactions with an unsupervised machine learning algorithm

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Abstract

1. This gap in knowledge regarding ecological interactions has prompted the development of various predictive approaches. Traditionally, ecological interactions have been inferred using traits. However, the lack of trait information for numerous organisms necessitates using phylogenetic data and statistical insights from interaction matrices for prediction. Previous studies have overlooked the validation of model-predicted interactions.
2. This study used a novel method in predicting ecological interactions using a self-organizing map (SOM), an unsupervised machine learning algorithm. The SOM learns from the input interaction matrix by grouping the nodes into output layers based on their interactions. Subsequently, the trained model predicts the interactions as scores. To distinguish between interactions and non-interactions, we employed F1 score maximization, setting scores above a specific threshold as interactions and the remainder as non-interactions. We applied this method to three unipartite metawebs and one bipartite metaweb and subsequently validated the predicted interactions using two innovative approaches: taxonomic and interaction recovery validation.
3. Our method exhibited outstanding predictive performance, particularly for large networks. Various binary classification performance indicators, including F1 score (0.84–0.97) and accuracy (0.97–0.99), indicated high performance. Moreover, the method generated minimal predicted interactions, signifying low noise in the predictions, particularly for large networks. Taxonomic validation excels in metawebs with a connectance >0.1 but performs poorly in metawebs with very low connectance. In contrast, interaction recovery was most effective in larger metawebs.
4. Our proposed method excels at making highly accurate predictions of ecological interactions with minimal noise, solely utilizing input interaction data without relying on traits or phylogenetic information regarding interacting nodes. These predictions are particularly precise for large networks, underscoring their potential to address knowledge gaps in emerging extensive metawebs. Notably, taxonomic validation and interaction recovery methods are sensitive to connectance

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and network size, respectively, suggesting prospects for developing robust interaction validation methods.

KEYWORDS

ecological interaction, ecological network, Eltonian shortfall, interaction prediction, interaction validation, metaweb, network prediction, self-organizing map (SOM)

1 | INTRODUCTION

Ecological interactions represent an intricate web of relationships that govern the dynamics of natural ecosystems (Ulanowicz, 2004). These interactions, which encompass predation, competition, mutualism, and more, are fundamental drivers that shape biodiversity, community structure, and ecosystem functioning (Tylianakis et al., 2008). Yet, unravelling ecosystem dynamics through ecological interactions encounters two significant challenges: (i) the scarcity of interaction data, often referred to as the 'Eltonian Shortfall' (Hortal et al., 2015) and (ii) the geographical bias in observed interactions, confined to specific biogeographic regions (Cameron et al., 2019; Poisot et al., 2023). This scarcity of data arises from the sheer complexity of potential interactions among a limited number of organisms, coupled with the substantial expertise and effort required to identify interacting species and characterize their interactions (Hortal et al., 2015; Jordano, 2016). Consequently, our understanding of ecological interactions has left observed ecological networks undersampled and incomplete. Bridging this knowledge gap requires the development of models to predict ecological interactions (Adhurya et al., 2023; Morales-Castilla et al., 2015).

Ecological interactions depend on several factors that progressively shape their patterns and dynamics. The co-presence of different interacting species within a specific spatiotemporal niche is determined by traits such as life history (Laughlin et al., 2012; White et al., 2007; Woodward et al., 2005), phenology (Encinas-Viso et al., 2012) and environmental tolerance (Lima-Mendez et al., 2015; Sunday et al., 2012). Subsequently, morphological (Ings et al., 2009; Rudolf, 2012) and behavioural traits (Pruitt & Ferrari, 2011) determine the potential for interaction. Furthermore, the number of individuals in different species defines the intensity of interactions (Olito & Fox, 2015; Pomeranz et al., 2019). Consequently, modelling ecological interactions often involves the consideration of species traits, commonly referred to as niche-based or trait-matching models (Bartomeus et al., 2016; Gravel et al., 2013; Williams & Martinez, 2000). However, niche models typically encompass just one or two traits, making it challenging to comprehend the full spectrum of traits that structure ecological networks, resulting in the 'intervality' problem and occasional erroneous predictions (Allesina et al., 2008; Cattin et al., 2004; Eklöf et al., 2013).

With the increasing availability of trait databases and extensive food web data, some studies have turned to machine learning to explore 'trait-matching' in ecological interactions using multiple traits (Desjardins-Proulx et al., 2017). Nonetheless, obtaining comprehensive trait information for interacting species remains challenging due to data scarcity across various taxa and geographic biases, known as

the 'Raunkiaeran shortfall' (Hortal et al., 2015). The 'intervality' issue and the scarcity of trait data have prompted a shift in our approach to modelling ecological interactions. Recognizing the co-evolution between interacting partners (Agrawal, 2001; Thompson, 1999), researchers have explored the use of phylogenetic data as a proxy for traits in predicting ecological interactions, demonstrating superior performance compared to traditional trait-based models (Brousseau et al., 2018; Elmasri et al., 2020; Pearse & Altermatt, 2013).

However, there also exists another kind of modelling approach where ecological interactions are solely predicted by understanding the pattern of the input interaction matrix mathematically or statistically. These approaches encompass the stochastic block model/group-based model (SBM) (Allesina & Pascual, 2009), latent trait model (LT) (Rohr et al., 2010), and matching-centrality model (MC) (Rohr et al., 2016). Unlike traditional niche models, these approaches operate without needing trait or phylogenetic data and have demonstrated superior predictive performance. In the case of SBM, the model posits that interacting species form distinct groups and that interactions occur between these groups. In contrast, LT analyses the interaction matrix patterns and predicts interactions using one or more proxy traits, termed 'latent traits'. One latent trait can comprehend the effects of more than one actual trait and can be statistically related to actual traits. Meanwhile, MC aims to enhance LT's performance by incorporating centrality measures and assigning interaction probabilities to each node. A detailed overview of the different models for network prediction can be found in a recent review (Adhurya et al., 2023).

The advent of the 'metaweb', often referred to as a regional ecological network, has given rise to the creation of vast ecological networks (Adhurya et al., 2024a; Albouy et al., 2019; Ho et al., 2022; Maiorano et al., 2020a). This necessitates the exploration of machine learning techniques for interaction matrix-based prediction. In this study, we propose an innovative approach to predict ecological interactions by harnessing the power of a self-organizing map (SOM) (Kohonen, 1982, 2001), also known as the Kohonen map, an unsupervised machine learning algorithm. SOM preserves the topological properties of input data, making it a valuable tool for exploring high-dimensional datasets, identifying patterns or groupings within them, and further projecting them to lower-dimensional space (Kohonen, 2001; Moniruzzaman et al., 2021; Park et al., 2018; Park, Céréghino, et al., 2003). Different ordination techniques (e.g. PCA, principal component analyses; MDS, multidimensional scaling; t-SNE, t-distributed stochastic neighbour embedding) also help to reduce data dimension and classification, but those are not model. The scores obtained from those techniques can be used as feature variables to develop a model that can be further used for prediction

(Strydom et al., 2021). In contrast, SOM is both a dimension reduction technique and a model that can be used further for prediction. In addition, SOM is a machine-learning method which can be useful for capturing complex patterns of ecological interactions in large ecological networks. Like phylogeny-based predictive models and SBM, our rationale for employing SOM to predict ecological interactions relies on the premise that species with similar interaction patterns often share common interaction partners.

This study has three primary objectives: (a) to establish a methodology that utilizes SOM to predict ecological interactions; (b) to evaluate the model's predictive performance using binary classification performance indicators and (c) to measure the validity of predicted links by introducing two novel methods.

2 | MATERIALS AND METHODS

2.1 | Predicting ecological interactions with SOM

Our approach for predicting ecological interactions using SOM begins with inputting an interaction matrix. The SOM operates by classifying the input data into output layers by adjusting the weight of the neurons (i.e. computational units) based on the distances

between the input data points. The SOM comprises an input layer representing the interaction matrix and an output layer (the output map) connected by weight vectors (Figure 1). The output layer typically consists of neurons arranged in a two-dimensional (2D) grid for effective visualization. The optimal arrangement is a hexagonal lattice, as it avoids favouring the horizontal and vertical directions found in the rectangular arrays. This 2D representation preserves the original topology of the input layer while reducing its dimensionality. For each input data point, the algorithm calculates the similarity between the input data and weight vectors of all neurons. The neuron with the weight vector closest to the input data becomes the 'best matching unit' (BMU). Weights of the BMU and its neighbouring neurons are updated to resemble the input data, thereby promoting clustering. As the training progresses, the learning rate decreases, leading to smaller weight adjustments and aiding convergence to a stable solution. The neighbourhood function determines which neurons are updated with the BMU, initially having a wide reach that gradually narrows as the training advances. Neurons within the neighbourhood are updated more than distant neurons. These steps are iterated over a predefined number of epochs, each time with a decreased learning rate and neighbourhood size. This process trains the neural network to classify the input vectors based on the weight vectors they closely resemble. Consequently,

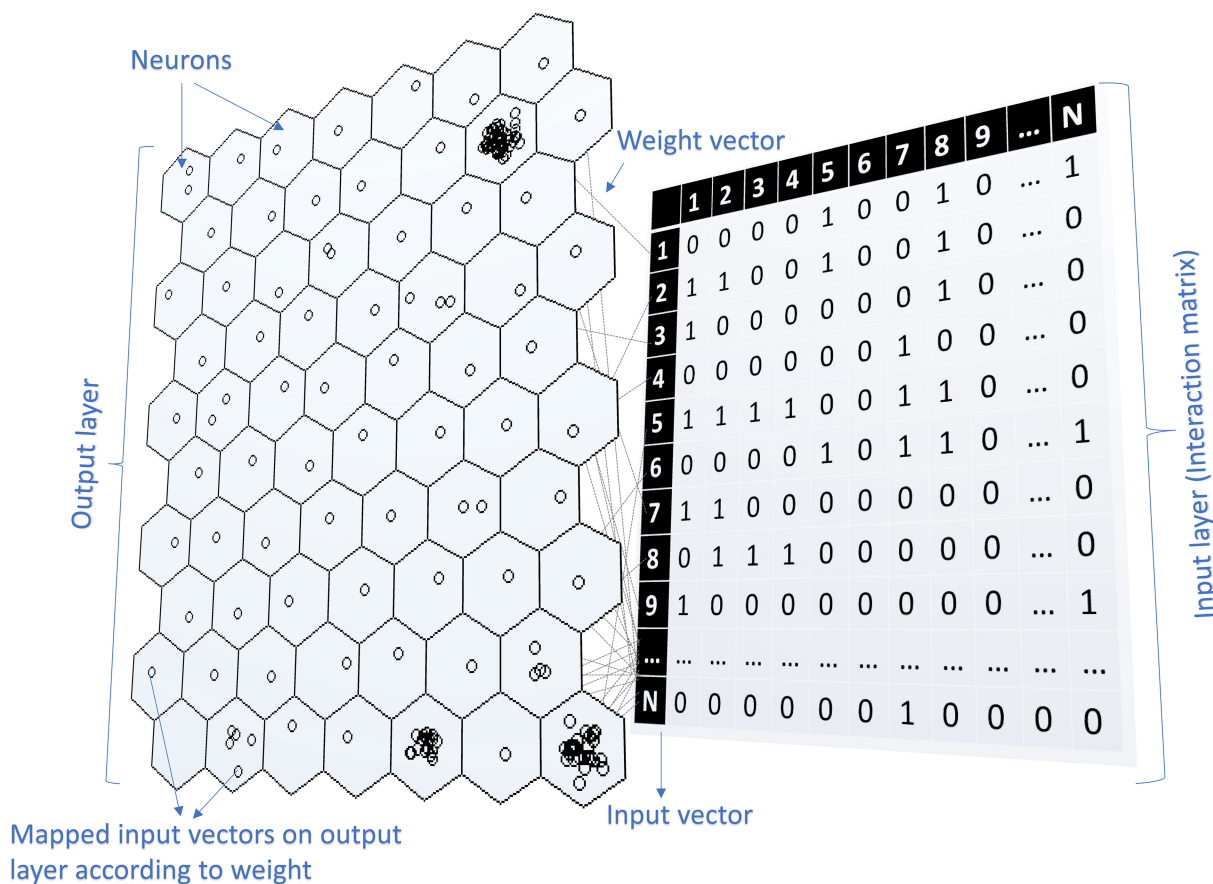


FIGURE 1 Illustration of self-organizing map (SOM) utilization for grouping interacting nodes according to their interactions. The SOM comprises an input layer containing the interaction matrix and an output layer consisting of neurons. Input vectors (nodes) are mapped onto the output layer using weight vectors determined by their interaction patterns.

after training, the SOM organizes the neurons in the grid to reflect the underlying structure of the input data. The detailed SOM algorithm for theoretical considerations can be found in the literature Kohonen (2001), and its ecological application can be found in many previous researchers (Chon et al., 1996; Park et al., 2014; Park, C  r  ghino, et al., 2003; Park, Chang, et al., 2003; Park, Lek, et al., 2006; Park, Tison, et al., 2006).

The selection of number of SOM output neurons, determined by dimension of the SOM map, plays a crucial role in revealing data deviations. If the map size is too small, significant differences that need to be identified may be overlooked. Conversely, an excessively large map may over-emphasize minor distinctions (C  r  ghino & Park, 2009). A heuristic guideline proposed by Vesanto and Alhoniemi (2000), also applied in the study of diatom communities (C  r  ghino & Park, 2009; Park, Tison, et al., 2006), recommends an optimal number of map units approximately equal to $5 \times \sqrt{n}$, where 'n' signifies the number of training samples (sample vectors). In this study, 'n' stands for the longer dimension of the input interaction matrix (if the dimension of the data is 150×100 , then $N = 150$). This approach involves computing the two largest eigenvalues of the training data, establishing the ratio between the side lengths of the map grid based on these eigenvalues, and adjusting the actual side lengths to approximate the number of map units defined by the $5 \times \sqrt{n}$ rule.

Given the binary nature of the interaction matrix used in this study, the Tanimoto coefficient (Jaccard, 1912; Tanimoto, 1958) was used for distance calculation. We adjusted three key hyperparameters: number of iterations, minimum learning rate and maximum learning rate. We monitored the training progress by assessing how the mean distance to the closest unit (BMU), also known as the quantization error (QE), changed during training. Subsequently, we increased the number of iterations and reevaluated the QE. If a decrease was observed, we continue to increase the number of iterations until no further decrease in QE occurs. We also checked whether changes in learning rates affected QE (Figure 2, Step 1). The predictions and validation procedures were executed using the R platform (R Core Team, 2023), leveraging the 'kohonen' package (Wehrens & Kruisselbrink, 2018) for SOM operations.

2.1.1 | Prediction and thresholding

Once a satisfactory model was achieved, we utilized the 'predict (model, data)' function of the 'stats' package (R Core Team, 2023) in R to generate predictions based on the input matrix. This function identifies the BMU for each input data point according to the trained model and returns the associated weight vector. These predictions produced scores ranging from 0 to 1, as binary matrix was used as input (Figure 2, Step 2). To establish a suitable threshold for distinguishing between interactions and non-interactions, we employed F1 scores (Canbek et al., 2022). Given the typical imbalance of ecological interaction matrices with numerous non-interactions (0), we

prioritized optimizing the threshold value to achieve the maximum F1 score, as it provides greater emphasis on positive interactions (1s) over non-interactions (0s) (Collell et al., 2018; Lipton et al., 2014). Our approach involved generating 1001 values between 0 and 1 at intervals of 0.001 and selecting the threshold that maximizes the F1 score. A predicted score greater than the threshold is considered as an interaction (1), and the rest are considered as non-interaction (0) (Figure 2, Step 3). To enhance computational efficiency, we employed 'future' (Bengtsson, 2021) and 'future.apply' (Bengtsson, 2021) packages to parallelize this task. The 'future' package allows to parallelize the process in the available cores of the operating device, while the 'future.apply' package allows to apply a particular function in the future package in parallel operation.

2.1.2 | Predictive performance evaluation

We assessed the overall model prediction by computing various binary classification model performance indicators, including the area under the receiver operating characteristic curve (ROC), accuracy, true skill statistic (TSS), and the proportion of true interactions (%) recovered (Canbek et al., 2022). ROC is calculated using the 'pROC' package (Robin et al., 2011) in R. However, we have already ensured the performance of the SOM by optimizing the QE to a minimum and the F1 score to the maximum during binary classification. Still, these additional performance measures allow us to not only check the overall performance of the model but also make a comparison with existing studies.

2.2 | Validation

The critical step of validating predicted ecological interactions has mostly been neglected in prior studies, thereby introducing uncertainties into the inferred links. Therefore, we propose two distinct validation approaches: (i) taxonomic validation and (ii) interaction recovery.

2.2.1 | Taxonomic validation

Since ecological interactions often evolve through reciprocal co-evolution, taxonomically similar organisms frequently exhibit similar ecological interactions (Agrawal, 2001; Thompson, 1999). Building on the same principle underpinning phylogeny-based predictive models, we introduced an innovative method for the taxonomic validation of our predicted interactions. Notably, no taxonomy-related information was provided during model training, ensuring the impartiality of our validation approach.

Figure 3 outlines the methodology, with several key steps. During validation, we first gathered taxonomic information for all nodes (Figure 3, Step 1). We assessed whether each predicted interacting partner interacted with another partner within the same taxonomic category in the observed network. If such interactions exist at lower

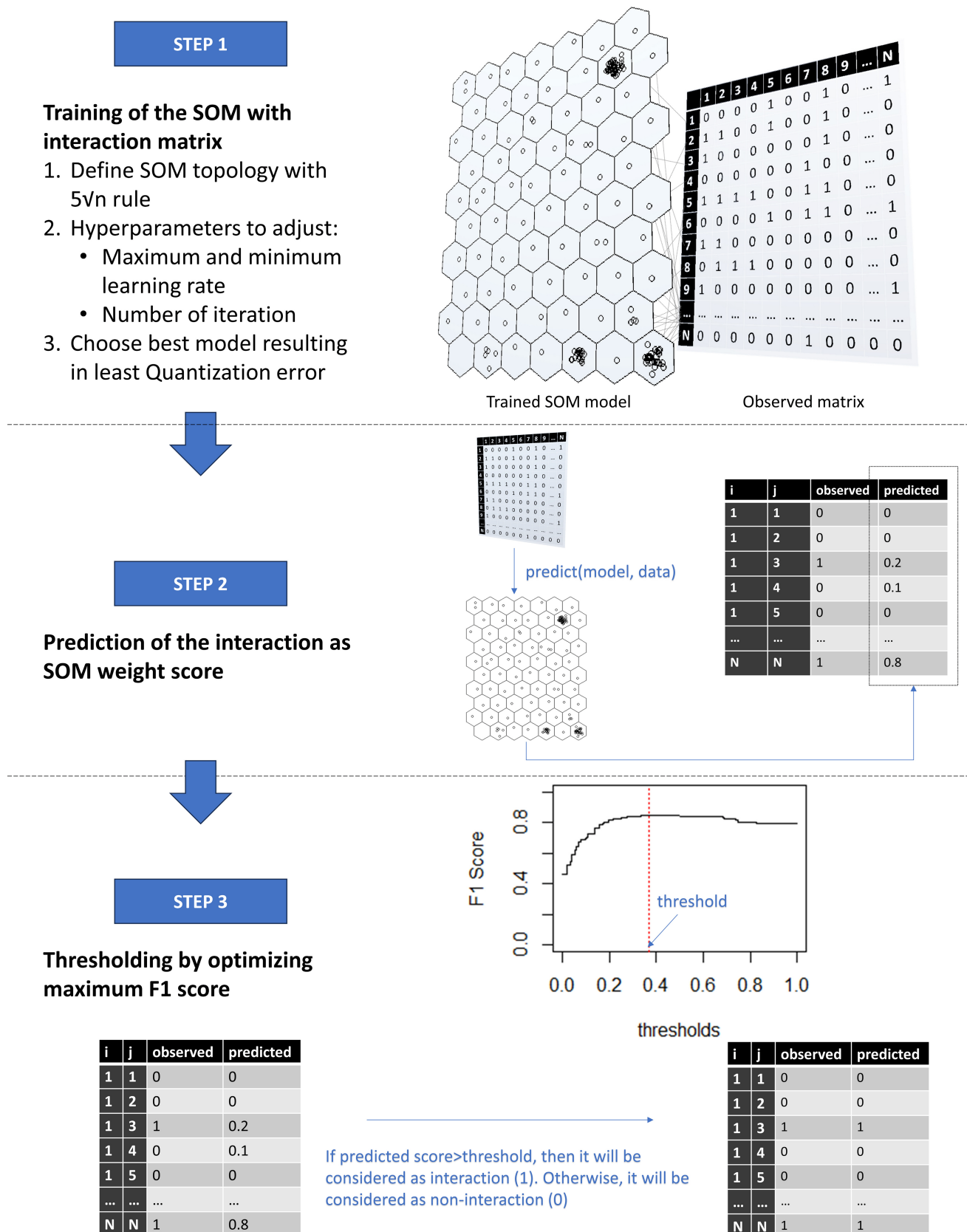


FIGURE 2 Conceptualization of the proposed methodology for predicting ecological interactions with self-organizing map (SOM).

taxonomic levels (e.g. tribe and family), we infer a high likelihood of interaction with the predicted partner. Conversely, if the predicted partner shares a taxonomic category with another partner at a higher

level (e.g. phylum), we infer a lower likelihood of that interaction. If no existing interacting partner shared a taxonomic category with the predicted partner, even at higher levels, we deemed this interaction

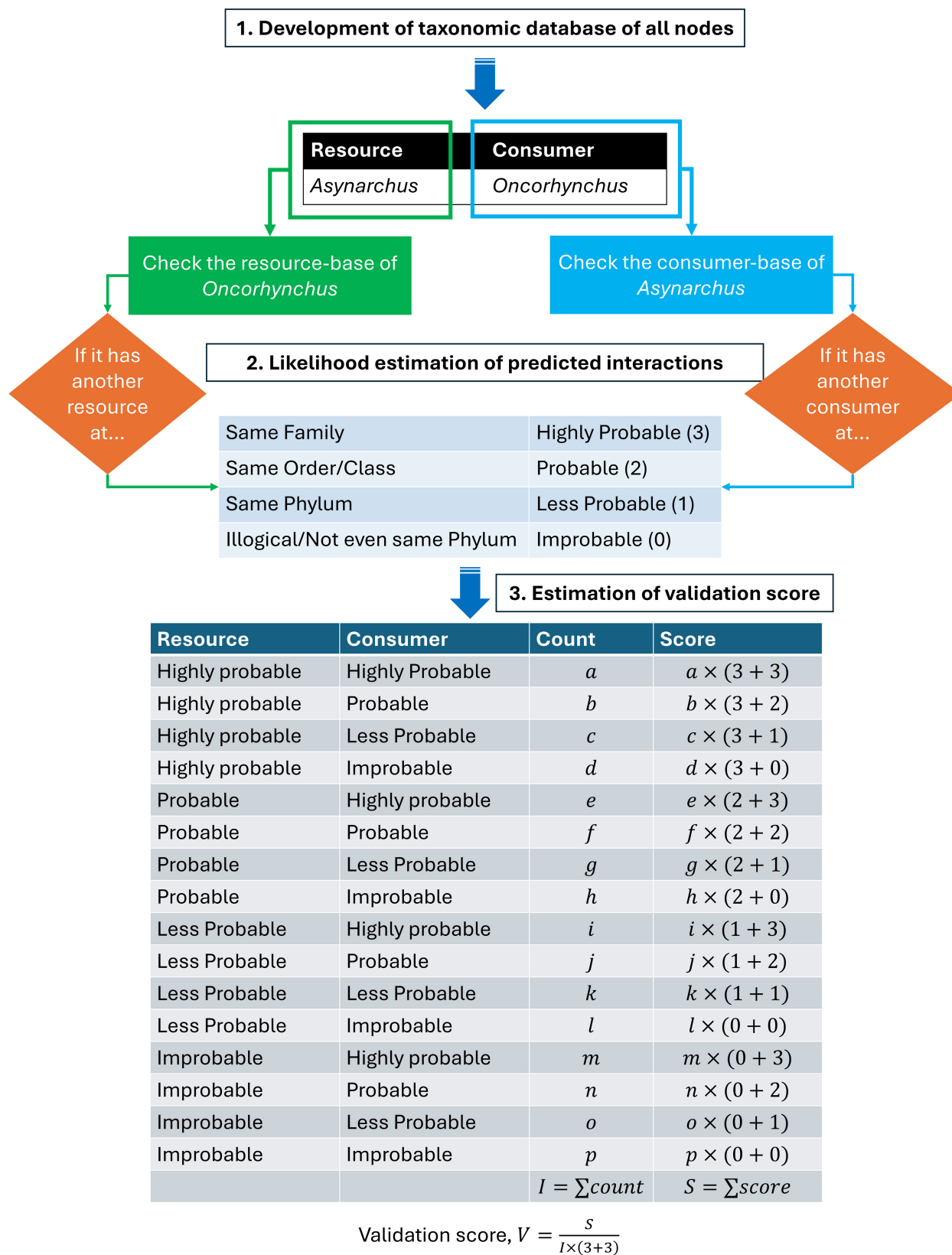


FIGURE 3 Flowchart indicating the procedure of taxonomic validation of predicted interactions. See Section 2.2.1 for the explanation.

'Improbable'. Subsequently, we attempted to establish a comprehensive validation measure. We assigned scores to the likelihood of correctly predicted interactions in a rank-wise manner: Improbable (0),

Less Probable (1), Probable (2), and Highly Probable (3), as illustrated in step 2 of Figure 3 and Table 1. Depending on the desired validation resolution, intermediate ranks can be introduced. Then, we tallied

the number of predicted links with different probability combinations and weighed them using the corresponding scores. The total score (S) was divided by the total number of predicted interactions (I) and multiplied by the maximum score to yield a comprehensive validation measure (V) (Figure 3, Step 3).

2.2.2 | Interaction recovery

We randomly removed 1%, 2%, 3%, 4% and 5% of the interactions from the network and trained our model using the remaining interactions, creating an undersampled observed network. We then examined the number of removed links recovered in the model's predictions. It should be noted that removing too many interactions drastically alters the structure of the interaction matrix, potentially leading to significantly different predictions. Interaction matrix-based predictive methods rely solely on matrix properties; therefore, we limited the removal to 5% to maintain matrix integrity. This approach is similar to investigating the niche model's sensitivity to sampling effort in a previous study (Gravel et al., 2013).

TABLE 1 Definition and explanation of different interaction likelihood terms used for taxonomic validation scoring.

Likelihood of interacting partner	Explanation
Highly probable	There are one or more interacting partners belongs to same lowest taxonomic category (e.g. genus or family) as the predicted partner
Probable	There are one or more interacting partners belongs to same higher taxonomic category (e.g. Order and Class) as the predicted partner, but not at the lowest taxonomic level
Less probable	There are one or more interacting partners belongs to same highest taxonomic category (e.g. Phylum) as the predicted partner, but there is no match at the lower taxonomic levels
Improbable	There is no match of the predicted partner with the existing interacting partners even at the highest taxonomic level

TABLE 2 Scoring system used for different metawebs to estimate taxonomic validation measures.

Likelihood of interacting partner (score)	Taxonomic category where matching was found				
	KF	TEU	BS	BAP-plant	BAP-ant
Highly probable (3)	Family	Genus/Family	Family	Genus	Genus
Probable (2)	Class/Order	Order	Class/Order	Family	Tribe
Less probable (1)	Phylum	Class	Phylum	Order	Subfamily
Improbable (0)	Not even at the highest taxonomic level				

Abbreviations: BAP, Brazilian Ant-Plant metaweb; BS, Barents Sea metaweb; KF, KF metaweb; TEU, Tetra-EU metaweb.

2.3 | Applications

Our methodology was applied to predict and validate the interactions within four distinct metawebs, providing a comprehensive assessment of its performance. The first three metawebs primarily represent trophic interactions and are unipartite in nature: the KF metaweb (KF) (Adhurya et al., 2024a), Tetra-EU (TEU) (Maiorano et al., 2020a), and Barents Sea (BS) metaweb (Planque et al., 2014). The KF encompasses 446 nodes (i.e. taxa) and 23,074 interactions, resulting in a connectance of 0.116; the TEU consists of 1151 nodes and 83,568 interactions, with a connectance of 0.063; and the BS metaweb encompasses 233 nodes and 2218 interactions, yielding a connectance of 0.04. In contrast, the Brazilian Ant-Plant metaweb (BAP) (Dáttilo & Vasconcelos, 2019) represents a bipartite mutualistic interaction network, featuring 2056 interactions between 178 plants and 101 ants with a connectance of 0.11. In case of bipartite network, the longer dimensions are used as row and shorter as column to increase the number of data to train the model.

Same hyperparameters were used for all metawebs during SOM training, with a minimum learning rate of 0.01, maximum learning rate of 1.5, and a fixed number of iterations at 1000. This uniform configuration enabled us to compare the performance of SOM across metawebs with varying structural properties. However, the hyperparameters can be further optimized by reducing the QE to obtain a better predictive performance. The SOM grid size is chosen according to $5 \times \sqrt{n}$ formula (KF: $105 = 7 \times 15$, TEU: $170 = 10 \times 17$, BS: $77 = 7 \times 11$ and BAP: $70 = 7 \times 10$). The random number generator in R used in this study was set to seed 123. For validation, the nodes in the KF and BS were classified into family, order, class and phylum; nodes in the TEU were classified into genus, family, order and class; plant nodes in the BAP were classified into genus, family and order; and ant nodes in the BAP were classified into genus, tribe and subfamily. Except for ants, we followed the GBIF backbone taxonomy (GBIF Secretariat, 2023) for the classification; the Catalogue of Life Taxonomy September 2023 checklist version (Bánki et al., 2023) was followed in the case of ants. The scoring systems for validation in the different metawebs are presented in Table 2. The source code for R scripts and supplementary files used in this study are available from the figshare data repository (Adhurya & Park, 2024).

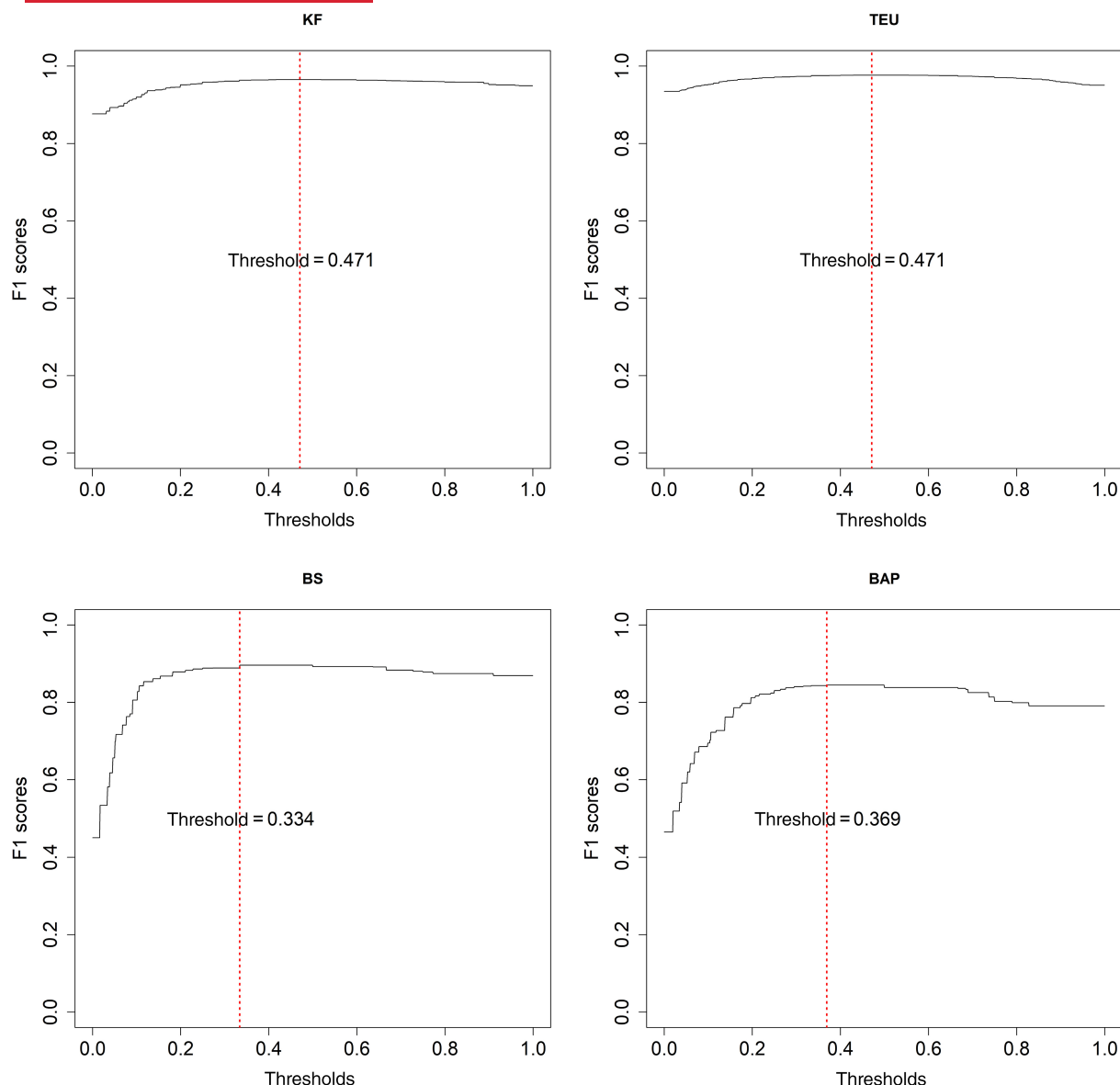


FIGURE 4 Threshold for defining predicted score as interaction (if the score > threshold) as obtained by maximizing F1 score in different metawebs. BAP, Brazilian Ant-Plant metaweb; BS, Barents Sea metaweb; KF, KF metaweb; TEU, Tetra-EU metaweb.

3 | RESULTS

3.1 | SOM training quality evaluation and thresholding

The QEs for the KF, TEU, BS and BAP decreased significantly from their untrained values of 0.02, 0.009, 0.019 and 0.037 to 0.003, 0.001, 0.005 and 0.013, respectively, after training. These reduced QEs indicate high-quality training of the SOM maps. Notably, there was a noticeable difference in threshold determination between the larger and smaller networks (Figure 4). Larger networks such as KF and TEU had a comparatively higher threshold (0.471), as determined by the maximization of the F1 score, in contrast to smaller networks with a threshold of 0.334.

3.2 | Prediction of interactions in four metawebs

The models predicted 457, 1730, 114 and 121 interactions in the KF, TEU, BS and BAP metawebs, respectively (Supporting Information: Predicted links and validation results.xlsx). Table 3 summarizes the predictive performance of our method across different networks, and the ROC (0.88–0.99) presented the high performance of the prediction models (Figure 5). Notably, our method excels in predicting interactions within large networks, such as KF and TEU, whereas its performance is somewhat less pronounced in smaller networks, such as BS and BAP. Accuracy, which assesses the correct prediction of both interactions and non-interactions, demonstrated a strong performance across all cases. However, when emphasizing interactions, as is evident in metrics such as %1 recovered, TSS and F1 scores, a

TABLE 3 Predictive performance of the model according to different binary classification measures for different metawebs used in this study.

Model performance measure	KF	TEU	BS	BAP
F1	0.97	0.98	0.89	0.84
ROC	0.98	0.99	0.93	0.88
TSS	0.95	0.97	0.85	0.77
Accuracy	0.99	0.99	0.99	0.97
%1 recovered	95.2	97.35	85.34	77.63
%TP	11.04	6.14	3.49	8.97
%TN	86.78	93.56	95.7	87.73
%FP (additionally predicted links)	0.17	0.13	0.21	0.72
%FN	0.56	0.17	0.60	2.58

Abbreviations: BAP, Brazilian Ant-Plant metaweb; BS, Barents Sea metaweb; FN, false negative; FP, false positive; KF, KF metaweb; TEU, Tetra-EU metaweb; TN, true negative; TP, true positive.

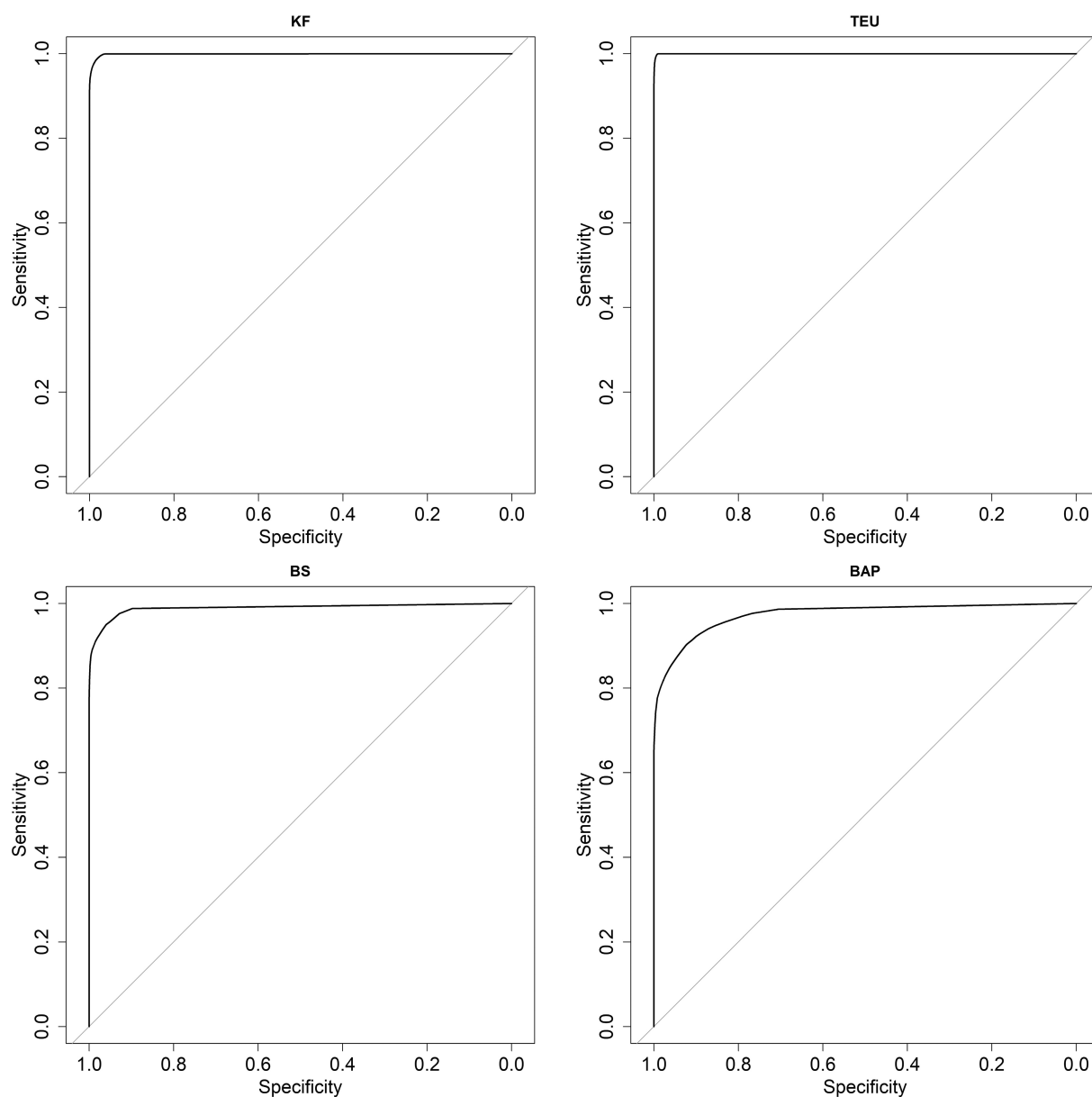


FIGURE 5 Model performance of the prediction models based on receiver operating characteristic curves (ROC) for different networks. BAP, Brazilian Ant-Plant metaweb; BS, Barents Sea metaweb; KF, KF metaweb; TEU, Tetra-EU metaweb.

distinct contrast emerges between large and small networks. This difference is primarily due to the increased number of false predictions in smaller networks.

Interestingly, our analysis indicates that the model performance remains unaffected by connectance. Furthermore, the proposed method exhibited minimal false predictions, except for the bipartite network BAP.

3.3 | Validation

3.3.1 | Taxonomic validation

The taxonomic validation method yielded diverse results when applied to the assessed networks (Table 4). KF and BAP exhibited strong validation performances, with scores of 0.745 and 0.73, respectively. Conversely, TEU and BS exhibit notably poorer performance, with validation scores of 0.21 and 0.12, respectively. These results suggest that the predicted links in the former case aligned with the phylogenetic relationships, whereas this alignment was absent in the latter two cases.

3.3.2 | Interaction recovery

The results of this method showed a clear positive relationship between the proportion of recovered links and network size. For example, in the case of the largest network (TEU), approximately 93.3% of the removed links were successfully recovered after eliminating 5% of

their interactions (4178 links). Conversely, the recovery of the removed links was notably limited in the case of small networks (Table 5).

All R scripts used in the study are available in [Supporting Information](#).

4 | DISCUSSION

4.1 | Comparison of the predictive performance with existing models

The overall performance of our proposed method demonstrates its effectiveness, particularly in the case of large networks. This suggests its suitability for predicting such networks with the added benefit of producing low levels of noise in predictions.

TABLE 5 Result of interaction validation with interaction recovery method.

Proportion (%) of links removed	Proportion of links recovered			
	KF	TEU	BS	BAP
1	0.814	0.937	0.227	0.000
2	0.768	0.935	0.136	0.000
3	0.753	0.932	0.224	0.016
4	0.757	0.929	0.135	0.024
5	0.733	0.933	0.144	0.029

Abbreviations: BAP, Brazilian Ant-Plant metaweb; BS, Barents Sea metaweb; KF, KF metaweb; TEU, Tetra-EU metaweb.

Partner likelihood					
Resource	Consumer	KF	TEU	BS	BAP
Highly probable	Highly probable	89	28	3	36
Highly probable	Probable	158	11	1	3
Highly probable	Less probable	0	23	0	4
Highly probable	Improbable	3	26	1	6
Probable	Highly probable	21	4	0	25
Probable	Probable	107	7	7	1
Probable	Less probable	0	12	0	4
Probable	Improbable	6	249	3	1
Less probable	Highly probable	0	0	0	18
Less probable	Probable	0	13	0	3
Less probable	Less probable	0	115	0	1
Less probable	Improbable	0	825	0	2
Improbable	Highly probable	22	4	0	12
Improbable	Probable	50	3	11	1
Improbable	Less probable	0	45	0	2
Improbable	Improbable	1	365	88	2
Total predicted interactions		457	1730	114	121

Abbreviations: BAP, Brazilian Ant-Plant metaweb; BS, Barents Sea metaweb; KF, KF metaweb; TEU, Tetra-EU metaweb.

TABLE 4 Count of different combinations of partner likelihood of the predicted interactions according to the taxonomic validation rules.

To assess the performance of our method comprehensively, we employed a range of binary classification model performance indicators. This approach allows for meaningful comparisons with existing ecological network prediction methods. For example, the niche model achieved TSS scores ranging from 0.13 to 0.76 across 11 small-sized food webs (Gravel et al., 2013). Another version of the niche model incorporating mechanistic formulations correctly predicted 5%–65% of the observed interactions (Petchey et al., 2008).

Among the interaction-matrix-based methods, the LT (Rohr et al., 2010) and MC (Rohr et al., 2016) models predicted 29%–87% and 50%–100% of the links in empirical webs, respectively. Another study (Terry & Lewis, 2020) compared the performances of stochastic block, LT, connectance (centrality) and MC models using ROC scores, resulting in scores of 0.691, 0.66, 0.676 and 0.671, respectively (Table 3).

The phylogenetic models achieved impressive ROC scores, with 0.93 obtained in the phylogeny-based generalized linear model (Pearse & Altermatt, 2013) and 0.917 in the latent-score network model (phylogeny-only model) (Elmasri et al., 2020). Another study (Brousseau et al., 2018) combined MC model (using real traits) and phylogenetic information to infer interactions, achieving a TSS of 0.65 and an accuracy of 83.4%.

Machine learning methods, such as k-nearest neighbour, demonstrated high predictive performance, with an ROC of 0.948 and 91.1% recovery (Elmasri et al., 2020). Another study utilized graph embedding following supervised machine learning with phylogenetic properties and predicted 92% of the observed interactions (Strydom et al., 2022).

Comparing these findings with those of our proposed model, it is evident that our method outperforms existing approaches across various performance measures. To facilitate more realistic model comparisons in the future, it will be imperative to apply all methods, particularly interaction matrix-based methods, to the same set of networks and evaluate them using all binary classification performance indicators.

4.2 | Validation

We devised two validation approaches: taxonomic validation and interaction recovery. Taxonomic validation yielded impressive scores for two networks (KF and BAP) while displaying poor performance for the other two (TEU and BS). It is important to note that the SOM relies solely on information provided during training. Therefore, similar predictions are expected if the observed interaction matrix lacks taxonomic bias. Our accurate predictive performance corroborated this finding. This indicated that taxonomically related organisms exhibited similar interactions in KF and BAP, whereas this was not the case for TEU and BS. This finding aligns with the accepted notion of ecological networks as 'ataxonomical aggregations' (Field et al., 1989).

Consequently, taxonomic validation might not be suitable for assessing ecological networks. This highlights the importance of cautiously assuming interactions at lower taxonomic levels based

on higher-level categories or functional groups during the metaweb development. Such assumptions can introduce an unreal taxonomic bias into the network, a common practice in developing networks from secondary data because of the unavailability of high-resolution data (Delmas et al., 2019; Hortal et al., 2015). This finding also prompted a re-evaluation of phylogeny-based models of metawebs such as TEU and BS. Finally, metawebs with higher validation scores tended to have a connectance above 0.1, suggesting that poor validation may also be linked to data scarcity within the metaweb.

Regarding the interaction recovery approach, we observed a size bias with larger networks exhibiting better validation performance than smaller networks, mirroring our predictive performance results. This implies that our method predicts the interactions within large networks, although it performs reasonably well for smaller networks.

It is important to note that the ultimate method for validating the predicted links confirms real-world interactions through empirical studies.

4.3 | Advantages and disadvantages of our methodology

Our methodology has significant advantages in terms of its ability to yield remarkably accurate predictions based solely on an interaction matrix. This is particularly beneficial for metawebs that encompass an extensive array of organisms, making it challenging to gather ecological trait information for such a vast number of nodes. Unipartite networks encompass a wide array of organisms with diverse traits and pose particular challenges. These networks make applying various types of trait information to different species challenging. For example, each group exhibits distinct traits in a trophic network encompassing microalgae, invertebrates and fish. This diversity limits the applicability of most trait-matching models to bipartite networks (Bartomeus et al., 2016; Elmasri et al., 2020; Pichler et al., 2020; Terry & Lewis, 2020; Weinstein & Graham, 2017). Although there are examples of trait matching in unipartite networks, they typically involve only one type of organism (Gravel et al., 2013; Pomeranz et al., 2019). However, there is an example of using a niche model for multiorganism unipartite networks (Petchey et al., 2008). When ecological trait data are lacking, approaches based on phylogenetic and interaction matrices offer viable alternatives. However, our analysis revealed that predicting interactions based on phylogeny becomes challenging if phylogenetically similar organisms do not share interaction partners. In such cases, matrix-based predictive methods may be more suitable for large unipartite networks comprising diverse organisms, even if the nodes are at different taxonomic levels. Therefore, our method has great potential for predicting interactions across various types of large networks, including unipartite, bipartite and k-partite networks.

However, the primary disadvantage of employing machine learning is its demand for a relatively large dataset, which leads to reduced performance in smaller networks. Another limitation of the

SOM is its challenge in elucidating the underlying reasons behind interactions despite providing robust and less noisy predictions compared with other methods. In contrast, while potentially less predictive, trait-based models offer valuable insights into the causal relationships driving interactions.

5 | CONCLUSIONS

This study introduces a novel method that employs unsupervised learning using the SOM algorithm to predict ecological interactions. The method exhibited robust predictive performance, particularly for large networks. However, this taxonomic validation approach may encounter challenges in networks with limited taxonomic dependencies within ecological interactions. The link recovery method underscores the effectiveness of the proposed method in accurately predicting links within large networks, albeit with limited success for smaller networks. This underscores their suitability for large-scale networks. In the future, this method should be improved to include a weighted matrix as the input. Moreover, this method can be tested with additional information about nodes (traits and phylogenetic distances) to compare its performance with other machine learning based method to efficiently utilize those information. Additionally, there is a research opportunity to delve deeper into the rationale behind the predicted interactions, akin to trait-based models, in which the influencing traits behind specific interactions can be identified. Moreover, there is room for further research to refine link validation methodologies because comprehending the actual predictions made by models remains critical in this field of study.

AUTHOR CONTRIBUTIONS

Young-Seuk Park conceived the idea and Sagar Adhurya designed the methodology. Sagar Adhurya performed all data curation, formal analysis, visualization, and validation and wrote the original draft. Young-Seuk Park was further involved in the fund acquisition, project administration, supervision, manuscript review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data used for the application of the study are publicly available: KF-metaweb data from the Dryad Digital Repository <https://doi.org/10.5061/dryad.cz8w9gjb> (Adhurya et al., 2024b), Tetra-EU data from

the Dryad <https://doi.org/10.5061/dryad.jm63xsj7b> (Maiorano et al., 2020b), Barents Sea data from the figshare <https://doi.org/10.6084/m9.figshare.c.3306768.v1> (Planque et al., 2016), and Brazilian Ant-Plant Metaweb data from the figshare <https://doi.org/10.6084/m9.figshare.5950306.v4> (Vasconcelos & Dáttilo, 2018). Source code for R scripts and supplementary files are available from the figshare <https://doi.org/10.6084/m9.figshare.24238930.v1> (Adhurya & Park, 2024).

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REFERENCES

- Adhurya, S., Agasti, N., & Park, Y.-S. (2023). Metaweb and its applications in understanding ecological interactions. *EcoEvoRxiv*.
- Adhurya, S., Lee, D.-Y., & Park, Y.-S. (2024a). KF-metaweb: A trophic metaweb of freshwater ecosystems of South Korea. *Global Ecology and Biogeography*, e13845. <https://doi.org/10.1111/geb.13845>
- Adhurya, S., Lee, D.-Y., & Park, Y.-S. (2024b). Data from: KF-metaweb: A trophic metaweb of freshwater ecosystems of South Korea (version 1.0) [dataset]. *Dryad Digital Repository*, <https://doi.org/10.5061/dryad.cz8w9gjb>
- Adhurya, S., & Park, Y.-S. (2024). Supplementary codes and files for the article 'A novel method for predicting ecological interactions with an unsupervised machine learning algorithm'. *Figshare*, <https://doi.org/10.6084/m9.figshare.24238930.v1>
- Agrawal, A. A. (2001). Phenotypic plasticity in the interactions and evolution of species. *Science*, 294, 321–326.
- Albouy, C., Archambault, P., Appeltans, W., Araújo, M. B., Beauchesne, D., Cazelles, K., Cirtwill, A. R., Fortin, M.-J., Galiana, N., Leroux, S. J., Pellissier, L., Poisot, T., Stouffer, D. B., Wood, S. A., & Gravel, D. (2019). The marine fish food web is globally connected. *Nature Ecology & Evolution*, 3, 1153–1161.
- Allesina, S., Alonso, D., & Pascual, M. (2008). A general model for food web structure. *Science*, 320, 658–661.
- Allesina, S., & Pascual, M. (2009). Food web models: A plea for groups. *Ecology Letters*, 12, 652–662.
- Bánki, O., Roskov, Y., Döring, M., Ower, G., Hernández Robles, D. R., Plata Corredor, C. A., Stjernaard Jeppesen, T., Örn, A., Vandepitte, L., Hobern, D., Schalk, P., DeWalt, R. E., Ma, K., Miller, J., Orrell, T., Aalbu, R., Abbott, J., Adlard, R., Adriaenssens, E. M., & Al, E. (2023). *Catalogue of life checklist (version 2023-09-14)*. Catalogue of Life.
- Bartomeus, I., Gravel, D., Tylaniakis, J. M., Aizen, M. A., Dickie, I. A., & Bernard-Verdier, M. (2016). A common framework for identifying linkage rules across different types of interactions. *Functional Ecology*, 30, 1894–1903.
- Bengtsson, H. (2021). A unifying framework for parallel and distributed processing in R using futures. *The R Journal*, 13, 208.
- Brousseau, P., Gravel, D., & Handa, I. T. (2018). Trait matching and phylogeny as predictors of predator–prey interactions involving ground beetles. *Functional Ecology*, 32, 192–202.
- Cameron, E. K., Sundqvist, M. K., Keith, S. A., CaraDonna, P. J., Mousing, E. A., Nilsson, K. A., Metcalfe, D. B., & Classen, A. T. (2019). Uneven global distribution of food web studies under climate change. *Ecosphere*, 10, e02645.
- Canbek, G., Taskaya Temizel, T., & Sagioglu, S. (2022). PTOPI: A comprehensive review, analysis, and knowledge representation of binary classification performance measures/metrics. *SN Computer Science*, 4, 13.
- Cattin, M.-F., Bersier, L.-F., Banašek-Richter, C., Baltensperger, R., & Gabriel, J.-P. (2004). Phylogenetic constraints and adaptation explain food-web structure. *Nature*, 427, 835–839.

- Céréghino, R., & Park, Y.-S. (2009). Review of the self-organizing map (SOM) approach in water resources: Commentary. *Environmental Modelling & Software*, 24, 945–947.
- Chon, T.-S., Park, Y. S., Moon, K. H., & Cha, E. Y. (1996). Patternizing communities by using an artificial neural network. *Ecological Modelling*, 90, 69–78.
- Collell, G., Prelec, D., & Patil, K. R. (2018). A simple plug-in bagging ensemble based on threshold-moving for classifying binary and multi-class imbalanced data. *Neurocomputing*, 275, 330–340.
- Dáttilo, W., & Vasconcelos, H. L. (2019). Macroecological patterns and correlates of ant-tree interaction networks in neotropical savannas. *Global Ecology and Biogeography*, 28, 1283–1294.
- Delmas, E., Besson, M., Brice, M.-H., Burkle, L. A., Dalla Riva, G. V., Fortin, M.-J., Gravel, D., Guimarães, P. R., Hembry, D. H., Newman, E. A., Olesen, J. M., Pires, M. M., Yeakel, J. D., & Poisot, T. (2019). Analysing ecological networks of species interactions. *Biological Reviews*, 94, 16–36.
- Desjardins-Proulx, P., Laigle, I., Poisot, T., & Gravel, D. (2017). Ecological interactions and the Netflix problem. *PeerJ*, 5, e3644.
- Eklöf, A., Jacob, U., Kopp, J., Bosch, J., Castro-Urgal, R., Chacoff, N. P., Dalsgaard, B., de Sassi, C., Galetti, M., Guimarães, P. R., Lomáscolo, S. B., Martín González, A. M., Pizo, M. A., Rader, R., Rodrigo, A., Tylianakis, J. M., Vázquez, D. P., & Allesina, S. (2013). The dimensionality of ecological networks. *Ecology Letters*, 16, 577–583.
- Elmasri, M., Farrell, M. J., Davies, T. J., & Stephens, D. A. (2020). A hierarchical Bayesian model for predicting ecological interactions using scaled evolutionary relationships. *The Annals of Applied Statistics*, 14, 221–240.
- Encinas-Viso, F., Revilla, T. A., & Etienne, R. S. (2012). Phenology drives mutualistic network structure and diversity. *Ecology Letters*, 15, 198–208.
- Field, J. G., Wulff, F., & Mann, K. H. (1989). The need to analyze ecological networks. In F. Wulff, J. G. Field, & K. H. Mann (Eds.), *Network analysis in marine ecology: Methods and applications* (pp. 3–12). Springer-Verlag.
- GBIF Secretariat. (2023). *GBIF backbone taxonomy*. Checklist dataset.
- Gravel, D., Poisot, T., Albouy, C., Velez, L., & Mouillot, D. (2013). Inferring food web structure from predator-prey body size relationships. *Methods in Ecology and Evolution*, 4, 1083–1090.
- Ho, H.-C., Brodersen, J., Gossner, M. M., Graham, C. H., Kaeser, S., Reji Chacko, M., Seehausen, O., Zimmermann, N. E., Pellissier, L., & Altermatt, F. (2022). Blue and green food webs respond differently to elevation and land use. *Nature Communications*, 13, 6415.
- Hortal, J., de Bello, F., Diniz-Filho, J. A. F., Lewinsohn, T. M., Lobo, J. M., & Ladle, R. J. (2015). Seven shortfalls that beset large-scale knowledge of biodiversity. *Annual Review of Ecology, Evolution, and Systematics*, 46, 523–549.
- Ings, T. C., Montoya, J. M., Bascompte, J., Blüthgen, N., Brown, L., Dormann, C. F., Edwards, F., Figueroa, D., Jacob, U., Jones, J. I., Lauridsen, R. B., Ledger, M. E., Lewis, H. M., Olesen, J. M., Van Veen, F. J. F., Warren, P. H., & Woodward, G. (2009). Review: Ecological networks—Beyond food webs. *Journal of Animal Ecology*, 78, 253–269.
- Jaccard, P. (1912). The distribution of the flora in the alpine zone. *New Phytologist*, 11, 37–50.
- Jordano, P. (2016). Sampling networks of ecological interactions. *Functional Ecology*, 30, 1883–1893.
- Kohonen, T. (1982). Self-organized formation of topologically correct feature maps. *Biological Cybernetics*, 43, 59–69.
- Kohonen, T. (2001). *Self-organizing maps* (3rd ed.). Springer Berlin Heidelberg.
- Laughlin, D. C., Joshi, C., van Bodegom, P. M., Bastow, Z. A., & Fulé, P. Z. (2012). A predictive model of community assembly that incorporates intraspecific trait variation. *Ecology Letters*, 15, 1291–1299.
- Lima-Mendez, G., Faust, K., Henry, N., Decelle, J., Colin, S., Carcillo, F., Chaffron, S., Ignacio-Espinosa, J. C., Roux, S., Vincent, F., Bittner, L., Darzi, Y., Wang, J., Audic, S., Berline, L., Bontempi, G., Cabello, A. M., Coppola, L., Cornejo-Castillo, F. M., ... Raes, J. (2015). Determinants of community structure in the global plankton interactome. *Science*, 348. <https://doi.org/10.1126/science.1262073>
- Lipton, Z. C., Elkan, C., & Naryanaswamy, B. (2014). Optimal thresholding of classifiers to maximize F1 measure. In T. Calders, F. Esposito, E. Hüllermeier, & R. Meo (Eds.), *Machine learning and knowledge discovery in databases* (pp. 225–239). Springer.
- Maiorano, L., Montemaggiore, A., Ficetola, G. F., O'Connor, L., & Thuiller, W. (2020a). TETRA-EU 1.0: A species-level trophic meta-web of European tetrapods. *Global Ecology and Biogeography*, 29, 1452–1457.
- Maiorano, L., Montemaggiore, A., Ficetola, G. F., O'Connor, L., & Thuiller, W. (2020b). Data from: Tetra-EU 1.0: A species-level trophic meta-web of European tetrapods [dataset]. *Dryad Digital Repository*, <https://doi.org/10.5061/dryad.jm63xsj7b>
- Moniruzzaman, M., Bhowmick, A. R., Karan, S., & Mukherjee, J. (2021). Spatial heterogeneity within habitat indicates the community assemblage pattern and life strategies. *Ecological Indicators*, 123, 107365.
- Morales-Castilla, I., Matias, M. G., Gravel, D., & Araújo, M. B. (2015). Inferring biotic interactions from proxies. *Trends in Ecology & Evolution*, 30, 347–356.
- Olito, C., & Fox, J. W. (2015). Species traits and abundances predict metrics of plant-pollinator network structure, but not pairwise interactions. *Oikos*, 124, 428–436.
- Park, Y.-S., Céréghino, R., Compin, A., & Lek, S. (2003). Applications of artificial neural networks for patterning and predicting aquatic insect species richness in running waters. *Ecological Modelling*, 160, 265–280.
- Park, Y.-S., Chang, J., Lek, S., Cao, W., & Brosse, S. (2003). Conservation strategies for endemic fish species threatened by the three gorges dam. *Conservation Biology*, 17, 1748–1758.
- Park, Y.-S., Chon, T.-S., Bae, M.-J., Kim, D.-H., & Lek, S. (2018). Multivariate data analysis by means of self-organizing maps. In F. Recknagel & W. K. Michener (Eds.), *Ecological informatics* (pp. 251–272). Springer International Publishing.
- Park, Y.-S., Kwon, Y.-S., Hwang, S.-J., & Park, S. (2014). Characterizing effects of landscape and morphometric factors on water quality of reservoirs using a self-organizing map. *Environmental Modelling & Software*, 55, 214–221.
- Park, Y.-S., Lek, S., Scardi, M., Verdonschot, P. F. M., & Jørgensen, S. E. (2006). Patterning exergy of benthic macroinvertebrate communities using self-organizing maps. *Ecological Modelling*, 195, 105–113.
- Park, Y.-S., Tison, J., Lek, S., Giraudel, J.-L., Coste, M., & Delmas, F. (2006). Application of a self-organizing map to select representative species in multivariate analysis: A case study determining diatom distribution patterns across France. *Ecological Informatics*, 1, 247–257.
- Pearse, I. S., & Altermatt, F. (2013). Predicting novel trophic interactions in a non-native world. *Ecology Letters*, 16, 1088–1094.
- Petchey, O. L., Beckerman, A. P., Riede, J. O., & Warren, P. H. (2008). Size, foraging, and food web structure. *Proceedings of the National Academy of Sciences of the United States of America*, 105, 4191–4196.
- Pichler, M., Boreux, V., Klein, A., Schleuning, M., & Hartig, F. (2020). Machine learning algorithms to infer trait-matching and predict species interactions in ecological networks. *Methods in Ecology and Evolution*, 11, 281–293.
- Planque, B., Primicerio, R., Michalsen, K., Aschan, M., Certain, G., Dalpadado, P., Gjøsæter, H., Hansen, C., Johannesen, E., Jørgensen, L. L., Kolsum, I., Kortsch, S., Leclerc, L.-M., Omli, L., Skern-Mauritzen, M., & Wiedmann, M. (2014). Who eats whom in the Barents Sea: A food web topology from plankton to whales. *Ecology*, 95, 1430.
- Planque, B., Primicerio, R., Michalsen, K., Aschan, M., Certain, G., Dalpadado, P., Gjøsæter, H., Hansen, C., Johannesen, E., Jørgensen, L. L., Kolsum, I., Kortsch, S., Leclerc, L.-M., Omli, L.,

- Stern-Mauritzen, M., & Wiedmann, M. (2016). *Who eats whom in the Barents Sea: A food web topology from plankton to whales*. Wiley. Collection. <https://doi.org/10.6084/m9.figshare.c.3306768.v1>
- Poisot, T., Ouellet, M.-A., Mollentze, N., Farrell, M. J., Becker, D. J., Brierley, L., Albery, G. F., Gibb, R. J., Seifert, S. N., & Carlson, C. J. (2023). Network embedding unveils the hidden interactions in the mammalian virome. *Patterns*, 4, 100738.
- Pomeranz, J. P. F., Thompson, R. M., Poisot, T., & Harding, J. S. (2019). Inferring predator-prey interactions in food webs. *Methods in Ecology and Evolution*, 10, 356–367.
- Pruitt, J. N., & Ferrari, M. C. O. (2011). Intraspecific trait variants determine the nature of interspecific interactions in a habitat-forming species. *Ecology*, 92, 1902–1908.
- R Core Team. (2023). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.
- Robin, X., Turck, N., Hainard, A., Tiberti, N., Lisacek, F., Sanchez, J.-C., & Müller, M. (2011). pROC: An open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics*, 12, 77.
- Rohr, R. P., Naisbit, R. E., Mazza, C., & Bersier, L.-F. (2016). Matching-centrality decomposition and the forecasting of new links in networks. *Proceedings of the Royal Society B: Biological Sciences*, 283, 20152702.
- Rohr, R. P., Scherer, H., Kehrl, P., Mazza, C., & Bersier, L. (2010). Modeling food webs: Exploring unexplained structure using latent traits. *The American Naturalist*, 176, 170–177.
- Rudolf, V. H. W. (2012). Seasonal shifts in predator body size diversity and trophic interactions in size-structured predator-prey systems. *Journal of Animal Ecology*, 81, 524–532.
- Strydom, T., Bouskila, S., Banville, F., Barros, C., Caron, D., Farrell, M. J., Fortin, M., Hemming, V., Mercier, B., Pollock, L. J., Runghen, R., Dalla Riva, G. V., & Poisot, T. (2022). Food web reconstruction through phylogenetic transfer of low-rank network representation. *Methods in Ecology and Evolution*, 13, 2838–2849.
- Strydom, T., Catchen, M. D., Banville, F., Caron, D., Dansereau, G., Desjardins-Proulx, P., Forero-Muñoz, N. R., Higinio, G., Mercier, B., Gonzalez, A., Gravel, D., Pollock, L., & Poisot, T. (2021). A roadmap towards predicting species interaction networks (across space and time). *Philosophical Transactions of the Royal Society, B: Biological Sciences*, 376, 20210063.
- Sunday, J. M., Bates, A. E., & Dulvy, N. K. (2012). Thermal tolerance and the global redistribution of animals. *Nature Climate Change*, 2, 686–690.
- Tanimoto, T. T. (1958). *An elementary mathematical theory of classification and prediction*. International Business Machines Corporation.
- Terry, J. C. D., & Lewis, O. T. (2020). Finding missing links in interaction networks. *Ecology*, 101, e03047.
- Thompson, J. N. (1999). The evolution of species interactions. *Science*, 284, 2116–2118.
- Tylianakis, J. M., Didham, R. K., Bascompte, J., & Wardle, D. A. (2008). Global change and species interactions in terrestrial ecosystems. *Ecology Letters*, 11, 1351–1363.
- Ulanowicz, R. E. (2004). Quantitative methods for ecological network analysis. *Computational Biology and Chemistry*, 28, 321–339.
- Vasconcelos, H., & Dáttilo, W. (2018). Ant_Plant_Networks_Savanna.csv. Figshare, <https://doi.org/10.6084/m9.figshare.5950306.v4>
- Vesanto, J., & Alhoniemi, E. (2000). Clustering of the self-organizing map. *IEEE Transactions on Neural Networks*, 11, 586–600.
- Wehrens, R., & Kruisselbrink, J. (2018). Flexible self-organizing maps in kohonen 3.0. *Journal of Statistical Software*, 87, 1–18.
- Weinstein, B. G., & Graham, C. H. (2017). On comparing traits and abundance for predicting species interactions with imperfect detection. *Food Webs*, 11, 17–25.
- White, E. P., Ernest, S. K. M., Kerkhoff, A. J., & Enquist, B. J. (2007). Relationships between body size and abundance in ecology. *Trends in Ecology & Evolution*, 22, 323–330.
- Williams, R. J., & Martinez, N. D. (2000). Simple rules yield complex food webs. *Nature*, 404, 180–183.
- Woodward, G., Ebenman, B., Emmerson, M., Montoya, J., Olesen, J., Valido, A., & Warren, P. (2005). Body size in ecological networks. *Trends in Ecology & Evolution*, 20, 402–409.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Data S1: Source code for R scripts and supplementary files.

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